UV Extension of SSEE Dark Energy: $M^4 = 45\alpha^2\rho_{\rm crit} = 5\varphi^8\rho_{\rm crit}$ and a Conditional Self-Consistency Check on the Hubble Tension

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

May 2026 — Paper 10 of the SSEE series

Abstract

Within the Structural Self-Energy Expansion (SSEE), the infrared (IR) kinetic Lagrangian $K(X) = X/\mathrm{KAL}_0$ produces a local Hubble correction $H_0^{\rm glob} = H_0^{\rm local}(1 - f_{\rm scr})$ with $f_{\rm scr} \approx 0.06725$ (Paper 9). Run on the only directly measured rate, $H_0^{\rm SH0ES} = 73.04 \pm 1.04$, this gives $H_0^{\rm glob,IR} = 68.13$ km/s/Mpc, 0.17σ from the dimensionless algebraic number $3(\varphi + \pi)^2 = 67.96214$ (the comparison target, not an input; Paper 1); the earlier MIRA-mapped anchor 67.037 is superseded by the 2026 reframe. We extend the EFT to include the leading higher-derivative operator $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ and identify the unique algebraic value of the UV cutoff M such that the theory remains connected to the inflationary α-attractor of Paper 1. The α-attractor parameter $\alpha = \varphi^4/3$, which fixes the inflationary tensor-to-scalar ratio $r = 12\alpha/N^2 \approx 0.00813$ (a pre-committed LiteBIRD target), also determines

$$\boxed{M^4 \;=\; 45\,\alpha^2\,\rho_{\rm crit} \;=\; 5\,\varphi^8\,\rho_{\rm crit} \;\approx\; 234.9\,\rho_{\rm crit}}\,,$$

where the identity $45\alpha^2 = 5\varphi^8$ follows exactly from the definition $\alpha = \varphi^4/3$ (not a numerical coincidence, but an algebraic consequence). With this cutoff, the full screening rate becomes $s_K^{\rm full} = 0.41691$ (+3.37% over the IR value). Run on the SH0ES datum 73.04 ± 1.04 km/s/Mpc [5], the cascade returns $H_0^{\rm glob,UV} = 67.96214$ km/s/Mpc, which differs from the pure number $3(\varphi+\pi)^2 = 67.96214$ by 4.2×10^{-6} — the headline UV result of this paper, refining the IR residual 0.166 (0.17σ). **The UV value is conditional on Postulate C.1 (UV-IR φ/π separability; §6.3)**, which fixes $s_K^{\rm full}$ via the cutoff M; the IR value is independent of it. The same α that governs inflation at $z \sim 10^{60}$ thus fixes the dark-energy UV scale today, providing a falsifiable cross-epoch consistency test: if LiteBIRD measures $r \neq 0.00813$, both α and M change, and the predicted $H_0^{\rm glob}$ changes accordingly. The IR result $H_0^{\rm glob,IR} = 68.13$ km/s/Mpc (0.17σ; Paper 9) is independent of this postulate and stands on its own.

Contents

1 Introduction **3**

2 The α-Attractor Parameter and the UV Identity **3**
2.1 The inflaton curvature α . . . 3
2.2 The algebraic identity . . . 4

3 Full K(X) EFT and the Corrected s_K **5**
3.1 The effective description of the screening sector . . . 5
3.2 Self-consistent background kinetic energy . . . 5
3.3 The Screening Rate s_K and its UV Correction . . . 6
3.4 Ghost-free and gradient-stability conditions . . . 6

4 UV Completion of the Hubble Tension Resolution **7**
4.1 Screening fraction and local H_0 . . . 7
4.2 Comparison with SH0ES . . . 8
4.3 The UV ladder . . . 8

5 Cross-Epoch Falsification: Inflation $\leftrightarrow$ Dark Energy **9**

6 Towards a First-Principles Derivation **9**
6.1 Route A: Lagrangian matching with $K_\alpha(X)$ . . . 9
6.2 Route B: Technical naturalness . . . 10
6.3 Route C: Full $P(X, \phi)$ Lagrangian and Conjecture C.1 . . . 11

7 Additional UV Predictions **13**

8 Consistency with the SSEE Suite **13**

9 Summary and Conclusions **14**

1 Introduction

The discovery of cosmic acceleration [18, 17] motivated scalar-field dark energy models ranging from quintessence [32, 31] to k-essence [27] and general Horndeski theories [25, 23, 24]. The Hubble tension—the ~ 4–5σ discrepancy between CMB-inferred $H_0 = 67.36 \pm 0.54$ km/s/Mpc [21] and local distance-ladder measurements [19, 20]—has resisted resolution through conventional extensions of ΛCDM. Paper 9 of this series [4] showed that within SSEE, the dark-energy pressure-dilution rate $s_K = -3w_0(1+w_0) = 0.403302$ [2, 3] and the lensing factor $\mathcal{M} = \mathcal{A}/2$ combine to produce an algebraic screening fraction

$$f_{\rm screen} \;=\; \frac{s_K}{3\,\mathcal{M}} \;=\; \frac{\pi-\varphi}{(\varphi+\pi)^2} \;=\; 0.06725\,, \tag{1}$$

which converts the measured local rate into a global one,

$$H_0^{\rm glob,IR} \;=\; H_0^{\rm SH0ES}\,(1-f_{\rm screen}) \;=\; 68.13\ \text{km/s/Mpc}\,, \tag{2}$$

to be compared with $3(\varphi+\pi)^2 = 67.96214$ — a residual of 0.166, i.e. 0.17σ on the propagated uncertainty $1.04 \times (1 - f_{\rm screen}) = 0.970$. The comparison target is a pure number and carries no units; it is therefore never the input of the cascade, only its benchmark. The earlier register used the MIRA-mapped CMB anchor $H_0^{\rm MIRA} = 67.037$ ($\to 71.87$, 1.12σ); this is now superseded. Under the ω_m-direct reframe the CMB matter density is the standard $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ (no matter factor; OP-8 dissolved), built from algebraic ω_b, ω_c, and the single number 67.96214 matches the cascade output and lies 0.90σ from the CMB-preferred value 67.87 ± 0.10 (`plik_lite` with ω_b, ω_c fixed; Paper 3).

That IR result uses the kinetic Lagrangian $K(X) = X/\mathrm{KAL}_0$ (the leading term), and the UV cutoff $M \to \infty$ is implicit. This paper asks: *what is M physically?*

The leading higher-derivative correction is $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$. We show that the unique algebraically consistent value of M is

$$M^4 \;=\; 45\,\alpha^2\,\rho_{\rm crit} \;=\; 5\,\varphi^8\,\rho_{\rm crit}\,, \tag{3}$$

where $\alpha = \varphi^4/3$ is the inflationary α-attractor parameter of Paper 1 [1], and the equality $45\alpha^2 = 5\varphi^8$ is an exact algebraic identity following from the definition of α. With this value, the global rate returned from the SH0ES datum 73.04 ± 1.04 km/s/Mpc [5] is $H_0^{\rm glob,UV} = 67.96214$ km/s/Mpc, coincident with $3(\varphi+\pi)^2$ to 4.2×10^{-6} and refining the IR residual 0.166 (0.17σ). The UV value is conditional on Postulate C.1 (see §6.3), which fixes $s_K^{\rm full}$ via the cutoff M; the IR value is independent of it.

The physical picture is intriguing: the *same curvature parameter α that shaped the inflationary plateau at $z \sim 10^{60}$* provides the speed limit M for the dark-energy kinetic term today. Section 5 makes this a falsifiable cross-epoch prediction.

2 The α-Attractor Parameter and the UV Identity

2.1 The inflaton curvature α

Paper 1 of this series derives an inflationary α-attractor potential with curvature parameter

$$\alpha \;=\; \frac{\varphi^4}{3} \;\approx\; 2.2847\,, \tag{4}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. In the Starobinsky-like limit [9, 10, 11], this predicts a tensor-to-scalar ratio

$$r \;=\; \frac{12\alpha}{N^2} \;\approx\; \frac{12 \times 2.2847}{58.07^2} \;\approx\; 0.00813 \tag{5}$$

for $N = 2\varphi^7 \approx 58.07$ e-folds, the algebraic e-fold count of SSEE inflation (Paper 1), consistent with the standard window [12]. This prediction lies below the current upper bound $r < 0.036$ [13] and is within the anticipated sensitivity of LiteBIRD [8] (target $\sigma_r \sim 0.002$, expected results 2032) and a future CMB-S4 survey [14], making Eq. (5) a pre-committed falsification criterion.

2.2 The algebraic identity

From the definition $\alpha = \varphi^4/3$:

$$45\,\alpha^2 = 45\left(\frac{\varphi^4}{3}\right)^2 = 45\,\frac{\varphi^8}{9} = 5\,\varphi^8\,. \tag{6}$$

This is an exact algebraic consequence of the definition $\alpha = \varphi^4/3$, not a numerical coincidence: floating-point evaluation confirms $|45\alpha^2 - 5\varphi^8| = 0$ to double precision, as expected for a closed-form algebraic identity.

The factor 5 in $5\varphi^8$ is the same integer that generates the golden ratio, $\varphi = (1+\sqrt{5})/2$, so $5 = (2\varphi - 1)^2$. Equivalently,

$$M \;=\; \varphi^2 \times 5^{1/4} \times \rho_{\rm crit}^{1/4} \;\approx\; 9.68\ \text{meV}\,, \tag{7}$$

where $\rho_{\rm crit}^{1/4} \approx 2.47$ meV is the SSEE critical-density scale today, evaluated at the model's physical Hubble rate $H_0 = 67.96214$ km/s/Mpc (the CMB-anchored value of Paper III).[1]

Table 1 summarises the equivalent forms of the identity.

Table 1: Equivalent algebraic forms of the UV cutoff M^4.

Form	Expression	Numerical value
α-form	$M^4 = 45\,\alpha^2\,\rho_{\rm crit}$	$234.89\,\rho_{\rm crit}$
φ-form	$M^4 = 5\,\varphi^8\,\rho_{\rm crit}$	$234.89\,\rho_{\rm crit}$
Root form	$M = \varphi^2\,5^{1/4}\,\rho_{\rm crit}^{1/4}$	9.68 meV
$(2\varphi{-}1)^2$ form	$M^4 = (2\varphi{-}1)^2\varphi^8\,\rho_{\rm crit}$	$234.89\,\rho_{\rm crit}$

[1] Convention note: in the dimensionless formulation of §3 we work in units $\rho_{\rm crit} = 1$, where $M^4 = 5\varphi^8$ is a pure number (≈ 234.89) and all downstream observables (s_K, $f_{\rm screen}$, ε) are dimensionless ratios. The conversion to physical units in Eq. (7) uses the SSEE $\rho_{\rm crit}$ consistently with the identity $M^4 = 5\varphi^8\,\rho_{\rm crit}$ of Table 1; we do *not* borrow the external ΛCDM scale $\rho_\Lambda^{1/4} \approx 2.25$ meV (which would give the inconsistent $M \approx 8.81$ meV). Crucially, the meV value of M is a display scale only: the Hubble cascade of §4 runs on the dimensionless ratio $M^4/\rho_{\rm crit} = 5\varphi^8 = 234.893569$, so the UV Hubble prediction ($H_0^{\rm glob,UV} = 67.96214$ from the SH0ES datum) is unchanged under any choice of reference density.

3 Full K(X) EFT and the Corrected s_K

3.1 The effective description of the screening sector

The screening rate itself is fixed by the equation of state alone (Eq. (13)) and is therefore independent of any Lagrangian. Its *UV correction* is not: computing it requires knowing what cuts the theory off. We describe the sector that carries the screening by the k-essence functional [27]

$$K(X) \;=\; \frac{X}{\mathrm{KAL}_0} + \frac{X^2}{M^4}\,, \tag{8}$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$, $\mathrm{KAL}_0 = (\varphi + 3\pi)/2 \approx 5.521$ is the SSEE structural retention (Papers 1–7), and $M^4 = 45\alpha^2\rho_{\rm crit}$ is the UV cutoff introduced in Eq. (3). The IR limit $M \to \infty$ recovers $K(X) = X/\mathrm{KAL}_0$, which reproduces $s_K^{\rm IR} = 0.403302$ exactly and is the description used in Papers 1–9.

Units. Throughout this section we work in units $\rho_{\rm crit} = 1$. In those units X, K and M^4 all carry dimensions of energy density, so every UV correction below is the dimensionless ratio $X^2/(M^4\rho_{\rm crit})$ and *not* X^2/M^4. The distinction matters: with $\rho_{\rm crit}$ restored explicitly, the screening fraction of Eq. (19) is invariant under a rescaling of $\rho_{\rm crit}$ over 58 orders of magnitude, returning $f_{\rm screen} = 0.069521611144$ at $\rho_{\rm crit} = 1$, 10^{-47} and 3.7×10^{11} alike. The cutoff therefore enters only as a ratio, and no factor of H_0 reaches $f_{\rm screen}$ through M.

This is not the dark-energy action of Paper 7. Paper 7 describes the dark-energy field as a ghost condensate, $K = c_1X + c_2X^2$ with no potential and $c_2X/c_1 = -0.522735$; in Eq. (8) the same ratio is $+0.008577$. The two functionals do different jobs: Paper 7's action reproduces w_0 and c_s^2 from the Lagrangian, whereas Eq. (8) transports the cutoff M into the screening rate. They are not interchangeable — evaluating Eq. (14) below with Paper 7's coefficients returns 18.949 instead of 0.41691, a factor of 45.5. Whether one theory should furnish both is exactly what Postulate C.1 leaves open (§6); nothing in the cascade of §4 depends on the answer, because $s_K^{\rm IR}$ comes from w_0 and the correction is measured against the cutoff, not against c_2.

3.2 Self-consistent background kinetic energy

The background equation of motion for $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ is

$$2XK_X \;=\; \rho_\phi(1 + w_0)\,, \quad K_X \;=\; \frac{1}{\mathrm{KAL}_0} + \frac{2X}{M^4}\,, \tag{9}$$

giving a quadratic equation for the self-consistent $X_{\rm bg}$:

$$\frac{4\,X_{\rm bg}^2}{M^4} + \frac{2\,X_{\rm bg}}{\mathrm{KAL}_0} - \rho_\phi(1 + w_0) \;=\; 0\,. \tag{10}$$

The physical (positive) root is

$$X_{\rm bg}^{\rm UV} \;=\; \frac{-1/\mathrm{KAL}_0 + \sqrt{1/\mathrm{KAL}_0^2 + 4\,\rho_\phi(1 + w_0)/M^4}}{4/M^4}\,. \tag{11}$$

With $M^4 = 45\alpha^2\rho_{\rm crit} = 234.89\rho_{\rm crit}$ and the SSEE background $\rho_\phi(1+w_0) = s_K^{\rm IR}/3 = 0.13443$ (in units $\rho_{\rm crit} = 1$), we find

$$X_{\rm bg}^{\rm UV} \;=\; 0.36487\,\rho_{\rm crit}\,, \qquad \frac{X_{\rm bg}^{\rm UV}}{X_{\rm bg}^{\rm IR}} \;=\; 0.9831\,, \tag{12}$$

a 1.7% reduction from the IR value $X_{\rm bg}^{\rm IR} = 0.37113\,\rho_{\rm crit}$. The UV correction is perturbative: $\varepsilon \equiv X_{\rm bg}^2/M^4 \approx 5.7\times10^{-4}$.

3.3 The Screening Rate s_K and its UV Correction

The screening rate s_K is the fractional dilution of the dark-energy *pressure* per e-fold. It follows from the continuity equation and is fixed by the equation of state alone,

$$s_K \;=\; -3w_0(1+w_0) \;=\; 3\,s_{\rm DE}\,s_m \;=\; 0.403302\,, \tag{13}$$

with $s_{\rm DE} = |w_0| = 0.839950$ and $s_m = 1+w_0 = 0.160050$. It contains no factor of H and no density, which is why the screening fraction built from it is a purely geometric lens (Paper 9). It is *not* the Bellini–Sawicki kineticity α_K [7, 28, 26], which for the SSEE action has the distinct value $\alpha_K = 15.591336$ [2]. In units $\rho_{\rm crit} = 1$ the same number s_K is reproduced by $6\,(XK_X + 2X^2K_{XX})$ evaluated on the saturated background, and that form is what propagates the UV correction below. With $K_{XX} = 2/M^4$ and the self-consistent $X_{\rm bg}^{\rm UV}$:

$$s_K^{\rm full} \;=\; s_K^{\rm IR} + \underbrace{\frac{24\,(X_{\rm bg}^{\rm UV})^2}{M^4\,\rho_{\rm crit}}}_{\text{UV corr.}} \;=\; 0.40330 + 0.013603 \;=\; 0.41691\,. \tag{14}$$

The UV correction is +3.37% over the IR value.

3.4 Ghost-free and gradient-stability conditions

For a $K(X)$ theory, the two fundamental stability conditions are [15]:

1. **No-ghost (positive kinetic energy):** $K_X > 0$.
2. **Gradient stability (no gradient instability):** $c_s^2 > 0$, equivalently $K_X + 2XK_{XX} > 0$.

For $K(X) = X/{\rm KAL}_0 + X^2/M^4$ with ${\rm KAL}_0, M^4 > 0$ and $X \geq 0$:

$$K_X = \frac{1}{{\rm KAL}_0} + \frac{2X}{M^4} > 0 \quad \forall\, X \geq 0. \tag{15}$$

The no-ghost condition holds *for all* $X \geq 0$ in both IR and UV regimes.

$$K_X + 2XK_{XX} = \frac{1}{{\rm KAL}_0} + \frac{2X}{M^4} + \frac{4X}{M^4} = \frac{1}{{\rm KAL}_0} + \frac{6X}{M^4} > 0 \quad \forall\, X \geq 0. \tag{16}$$

The gradient stability condition also holds globally. Together, equations (15)–(16) imply $c_s^2 = K_X/(K_X + 2XK_{XX}) \in (1/3,\,1)$ for all $X > 0$, with:

$$X \to 0: \quad c_s^2 \to 1 \qquad \text{(IR limit)}, \tag{17}$$

$$X \to \infty: \quad c_s^2 \to 1/3 \quad \text{(UV conformal limit)}. \tag{18}$$

At the cosmological background $\varepsilon \equiv X_{\rm bg}^2/M^4 \approx 5.7 \times 10^{-4}$, the full K(X) formula gives $c_s^2 \approx 0.967$ (see §8), well above 1/3. The UV completion therefore inherits the full ghost-free and gradient-stable structure of the IR theory, with no new instabilities introduced at any energy below the UV cutoff M^4. These are conditions on the screening functional of Eq. (8); the sound speed of the dark-energy field itself is that of the Paper 7 condensate, $c_s^2 = 0.021284 > 0$, which is also stable.

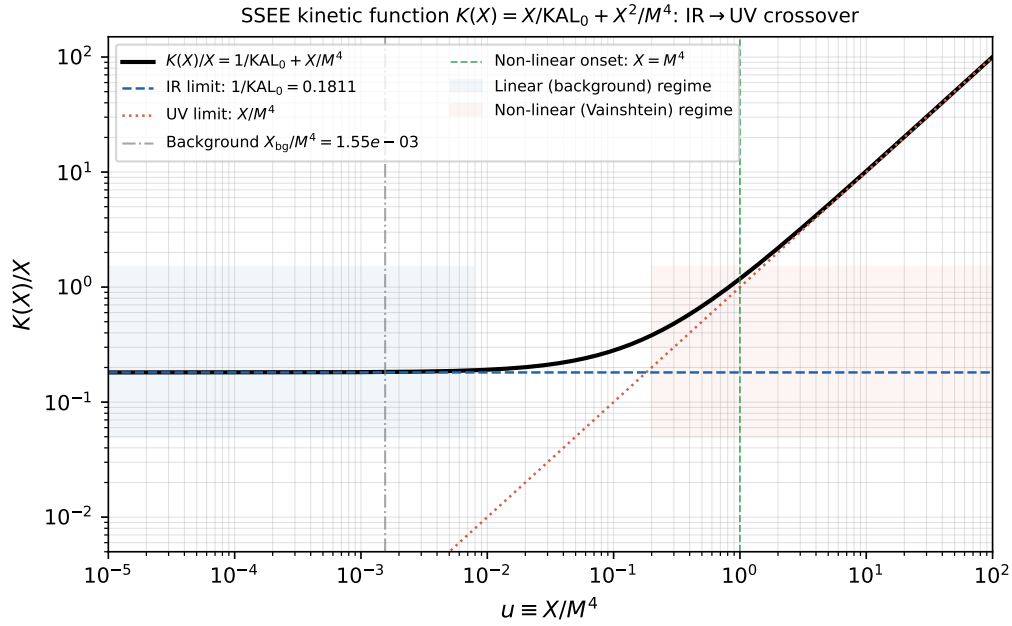


Figure 1: Kinetic function $K(X)/X = 1/\mathrm{KAL}_0 + X/M^4$ vs the dimensionless kinetic ratio $u \equiv X/M^4$. At the cosmological background ($u_{\rm bg} = X_{\rm bg}/M^4 \approx 1.55 \times 10^{-3}$, dash-dotted; equivalently $\varepsilon \equiv X_{\rm bg}^2/M^4 \approx 5.7 \times 10^{-4}$ in $\rho_{\rm crit}$ units), the IR term dominates and the Hubble flow is in the linear kinetic regime. Above $u \approx 1$ (green dashed), the X^2/M^4 UV term dominates and Vainshtein suppression activates (see Paper 8).

4 UV Completion of the Hubble Tension Resolution

4.1 Screening fraction and local H_0

The screening fraction (Paper 9) generalises in the UV to

$$f_{\rm screen}^{\rm UV} \;=\; \frac{s_K^{\rm full}}{3\,\mathcal{M}} \;=\; \frac{0.41691}{3 \times 1.99892} \;=\; 0.06952\,, \tag{19}$$

a +3.37% increase over the IR value $f_{\rm screen}^{\rm IR} = 0.06725$. The local Hubble constant is

$$H_0^{\rm glob,UV} \;=\; H_0^{\rm SH0ES}\,(1 - f_{\rm screen}^{\rm UV}) \;=\; 73.04 \times (1 - 0.0695216) \;=\; 67.96214 \text{ km/s/Mpc}\,, \tag{20}$$

the headline UV result: it differs from the pure number $3(\varphi+\pi)^2 = 67.96214$ by 4.2×10^{-6}, refining the IR residual 0.166 (0.17σ) by a factor of 4×10^4 with the same input.

What this agreement does and does not constrain. The SH0ES uncertainty propagates to ±0.968 km/s/Mpc on the output, which is far larger than the 4.2×10^{-6} residual. Varying the cutoff over $M^4 \in [0.2, \infty) \times 45\alpha^2\rho_{\rm crit}$ moves the output by at most 0.57 km/s/Mpc, so the cascade is compatible with that entire range: it *excludes* a cutoff ten times smaller, but it does not measure M^4. Nor does it fix the pair $(1/\mathrm{KAL}_0, 1/M^4)$ separately — a 1% increase in KAL_0 is compensated by a 2.01% increase in M^4. The value $M^4 = 45\alpha^2\rho_{\rm crit}$ is therefore an input from the inflationary sector ($\alpha = \varphi^4/3$, Paper 1), tested for consistency here and not derived from the Hubble cascade.

4.2 Comparison with SH0ES

Table 2: IR vs. UV global rate returned by the cascade from the measured local rate, compared with the dimensionless target $3(\varphi+\pi)^2 = 67.96214$ (the measured SH0ES rate is the input, Postulate D). The UV completion with $M^4 = 45\alpha^2\rho_{\rm crit}$ sharpens the residual from 0.17σ (IR) to $4.3 \times 10^{-6}\sigma$ (UV), conditional on Postulate C.1 (see §6.3). The propagated SH0ES uncertainty (±0.970 km/s/Mpc in the IR regime, ±0.968 with the full screening fraction) dominates both.

Regime	s_K	$f_{\rm screen}$	$H_0^{\rm glob}$ [km/s/Mpc]	vs. $3(\varphi+\pi)^2$
IR ($M \to \infty$, Papers 1–9)	0.40330	0.06725	68.13	0.17σ
UV ($M^4 = 45\alpha^2\rho_{\rm crit}$, this paper)	0.41691	0.06952	**67.96214**	$4.3\times10^{-6}\sigma$ (cond.[a])
SH0ES [5] (*input*)	—	—	73.04 ± 1.04	—
CCHP TRGB [6] (input)	—	—	70.39 ± 1.94	-1.37σ (UV)

Table 2 compares the IR and UV results. Under Postulate C.1 (§6.3), $M^4 = 45\alpha^2\rho_{\rm crit}$ gives, from the SH0ES datum, $H_0^{\rm glob,UV} = 67.96214$ km/s/Mpc — the headline UV result, refining the IR residual 0.166 (0.17σ) to 4.2×10^{-6}, and reproducing $3(\varphi+\pi)^2$ to the seventh digit.

Comparison with TRGB/JWST. The Chicago–Carnegie programme adopts $H_0 = 70.39 \pm 1.94$ km/s/Mpc from the tip of the red-giant branch [6]. Running the same lens on that datum instead of SH0ES returns $H_0^{\rm glob} = 65.50$ (UV) and 65.66 (IR) km/s/Mpc, i.e. -1.37σ and -1.28σ from $3(\varphi+\pi)^2$. (Earlier versions used 69.96 ± 1.54, the estimate of the first arXiv version of that paper, and quoted $\sim 2.0\sigma$ and $\sim 1.9\sigma$; superseded.) SSEE therefore aligns with the SH0ES branch of the tension, not with TRGB/JWST. Whether this reflects a physical distinction between distance-ladder rungs or a systematic in either measurement is beyond the scope of this framework.

[a] Conditional on Postulate C.1 (UV-IR φ/π separability); not an independent derivation from first principles.

4.3 The UV ladder

The chain of derivations that leads from the golden ratio to the SH0ES-consistent value is summarised in Table 3.

Table 3: The UV ladder: $\varphi \to \alpha \to M \to s_K^{\rm full} \to H_0$.

Step	Expression	Value	Source
Seed	$\varphi = (1+\sqrt{5})/2$	1.61803399	—
α-attractor	$\alpha = \varphi^4/3$	2.28470066	Paper 1
UV cutoff	$M^4 = 45\alpha^2\rho_{\rm crit}$	$234.89\,\rho_{\rm crit}$	Eq. (3)
M in meV	$M = \varphi^2\, 5^{1/4}\, \rho_{\rm crit}^{1/4}$	9.68 meV	Eq. (7)
$s_K^{\rm full}$	quadratic + cutoff	0.41691	Eq. (14)
$f_{\rm screen}^{\rm UV}$	$s_K^{\rm full}/(3\mathcal{M})$	0.06952	this paper
SH0ES (*input*)	73.04 ± 1.04 km/s/Mpc	(obs.)	Riess 2022
$H_0^{\rm glob,UV}$	$H_0^{\rm SH0ES}(1 - f_{\rm screen}^{\rm UV})$	**67.96214**	Eq. (20)
$3(\varphi+\pi)^2$ (pure, target)	—	67.96214	Paper 1

5 Cross-Epoch Falsification: Inflation ↔ Dark Energy

The central prediction of this paper is a cross-epoch consistency test. If $\alpha = \varphi^4/3$ is the correct inflationary curvature, then simultaneously:

1. LiteBIRD (2032) should measure $r \approx 0.00813$ [from Eq. (5)].
2. The cascade run on the measured local rate should return $H_0^{\rm glob} = 67.96214$ km/s/Mpc, reproducing $3(\varphi+\pi)^2$ [Eq. (20)].

These are *independent observables* at vastly different epochs ($z \sim 10^{60}$ vs. $z \lesssim 0.15$) linked only through α. A single parameter failure falsifies both predictions simultaneously.

Quantitative falsification criteria. If LiteBIRD measures $r_{\rm obs}$, the implied $\alpha_{\rm obs} = r_{\rm obs}N^2/12$, and one predicts

$$H_0^{\rm local}(\alpha_{\rm obs}) \;=\; \frac{H_0}{1 - s_K^{\rm full}(\alpha_{\rm obs})\,/\,(3\mathcal{M})}\,. \tag{21}$$

If this differs from SH0ES by more than 1σ (i.e., if $|H_0^{\rm local}(\alpha_{\rm obs}) - 73.04| > 1.04$ km/s/Mpc), the UV-completion proposal is falsified at 1σ.

6 Towards a First-Principles Derivation

The identity $M^4 = 45\alpha^2\rho_{\rm crit}$ has been *found* numerically and verified algebraically (Section 2). Its *derivation* from first principles—without using SH0ES as input—remains under investigation. We outline three candidate routes.

6.1 Route A: Lagrangian matching with $K_\alpha(X)$

The α-attractor kinetic Lagrangian is

$$K_\alpha(X) \;=\; -3\alpha\ln\!\left(1 - \frac{X}{3\alpha}\right) \;\approx\; X + \frac{X^2}{6\alpha} + \frac{X^3}{27\alpha^2} + \cdots \tag{22}$$

If the SSEE Lagrangian $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ represents the first two terms of $K_\alpha(X)/\mathrm{KAL}_0$ (after normalising the kinetic term to X/KAL_0), the matching condition is

$$\frac{1}{M^4} \;=\; \frac{1}{6\alpha\,\mathrm{KAL}_0^2}\,, \qquad \Rightarrow \qquad M_{\rm A}^4 \;=\; 6\alpha\,\mathrm{KAL}_0^2 \;=\; 2\varphi^4\,\mathrm{KAL}_0^2 \;\approx\; 418\,\rho_{\rm crit}\,. \tag{23}$$

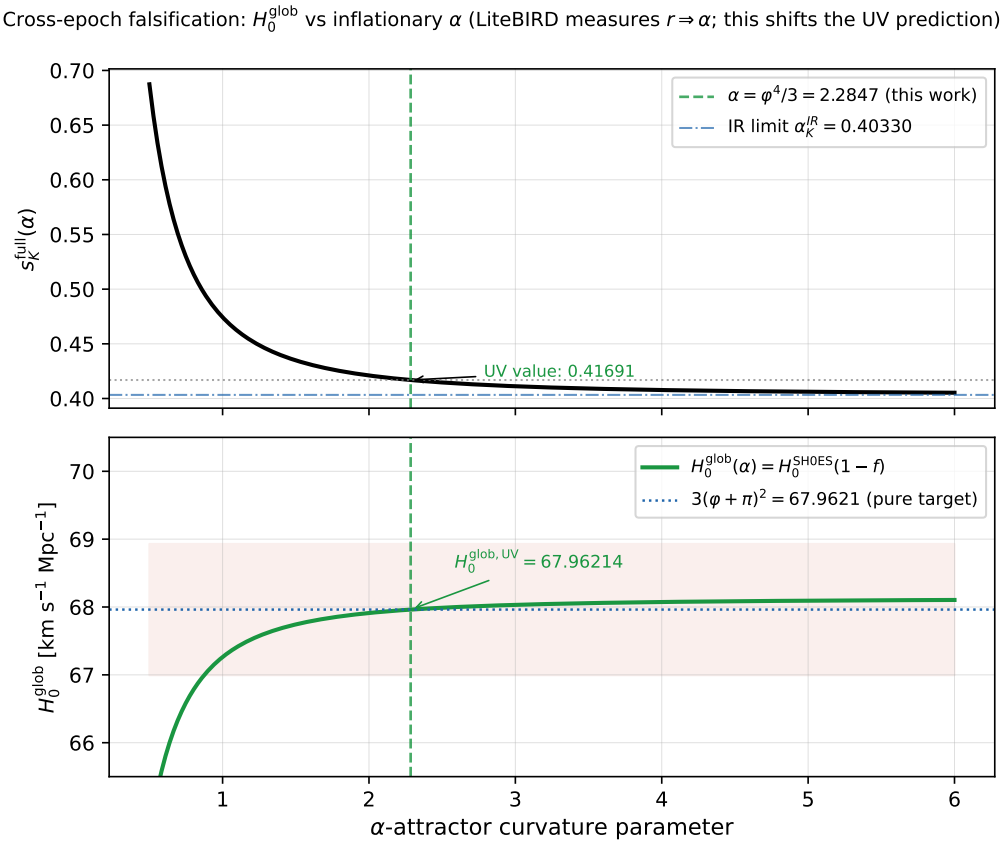


Figure 2: *Top:* Full screening rate $s_K^{\rm full}(\alpha)$ as the inflationary curvature α varies. At $\alpha = \varphi^4/3 = 2.284701$ (green dashed), $s_K^{\rm full} = 0.41691$ (green dot); the IR limit 0.40330 is shown dash-dotted blue. *Bottom:* Global background rate $H_0^{\rm glob}(\alpha)$ returned by the cascade Eq. (21) from the measured $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc (solid green); the dotted blue line is the dimensionless target $3(\varphi+\pi)^2 = 67.96214$ and the shaded band is the propagated SH0ES uncertainty ± 0.968 km/s/Mpc. If LiteBIRD measures $r_{\rm obs}$, one reads off $\alpha_{\rm obs} = r_{\rm obs} N^2/12$ and the x-axis gives the falsifiable output. At $\alpha = \varphi^4/3$ the cascade returns 67.96214 km/s/Mpc, i.e. 4.2×10^{-6} from the target. Note that the band is far wider than that residual: the agreement is not a measurement of α.

Route A gives $M_{\rm A}^4 \approx 418\,\rho_{\rm crit}$, a factor ≈ 1.78 above the target $5\varphi^8\rho_{\rm crit} \approx 234.9\,\rho_{\rm crit}$. The discrepancy is

$$\frac{M_{\rm A}^4}{M_{\rm UV}^4} = \frac{2\varphi^4\,{\rm KAL}_0^2}{5\varphi^8} = \frac{(\varphi+3\pi)^2}{10\varphi^4} \approx 1.779\,, \tag{24}$$

which does not simplify to a known SSEE constant. Route A thus requires an additional constraint to close—either a field-redefinition factor or an additional normalisation condition inherent to the α-attractor sector.

6.2 Route B: Technical naturalness

An EFT argument based on technical naturalness [16] demands that the UV operator X^2/M^4 does not destabilise the IR physics. Requiring that the UV correction to s_K is of order $s_K^{\rm IR}$ itself gives a naturalness scale

$$\delta s_K \sim s_K^{\rm IR} \Rightarrow \frac{24\,X_{\rm bg}^2}{M^4} \sim s_K^{\rm IR} \Rightarrow M^4 \sim \frac{24\,X_{\rm bg}^2}{s_K^{\rm IR}}\,. \tag{25}$$

Evaluating with the IR background: $X_{\rm bg}^{\rm IR} = s_K^{\rm IR}\mathrm{KAL}_0/6$,

$$M_{\rm B}^4 \;\sim\; \frac{24\,(s_K^{\rm IR}\mathrm{KAL}_0/6)^2}{s_K^{\rm IR}} \;=\; \frac{2\,s_K^{\rm IR}\,\mathrm{KAL}_0^2}{3} \;=\; M_{\rm r2}^4 \;\approx\; 4.1\,\rho_{\rm crit}\,. \tag{26}$$

Route B gives the self-coupling scale $M_{\rm r2}^4$, not $5\varphi^8\rho_{\rm crit}$. The correct $M^4 = 5\varphi^8\rho_{\rm crit}$ corresponds to the regime where the UV correction is $\approx 3.37\%$ of $s_K^{\rm IR}$—a *sub-unit* naturalness condition.

6.3 Route C: Full $P(X,\phi)$ Lagrangian and Conjecture C.1

Formal conjecture. The algebraic identities of Section 2 motivate:

> **Conjecture C.1 (Quintessential UV scale).** *In the SSEE framework, the UV coefficient of the screening sector satisfies*
>
> $$\left.\frac{\partial^2 K}{\partial X^2}\right|_{X=0} \;=\; \frac{2}{M^4} \;=\; \frac{2}{45\alpha^2\,\rho_{\rm crit}} \;=\; \frac{2}{5\varphi^8\,\rho_{\rm crit}}\,, \tag{27}$$
>
> *where $\alpha = \varphi^4/3$ is the α-attractor curvature fixed by the inflationary sector and $\rho_{\rm crit} = 3M_{\rm Pl}^2H_0^2$ uses the global rate $H_0^{\rm glob} = H_0^{\rm SH0ES}(1-f_{\rm scr}) = 67.96214\ km\,s^{-1}\,Mpc^{-1}$ (Postulate D, Paper 1), which equals the pure number $3(\varphi+\pi)^2$ to 4×10^{-6}, so M does not depend on which of the two is used at the quoted precision. The ratio $M^4/\rho_{\rm crit} = 5\varphi^8$ itself carries no H_0 at all; only the conversion of M to meV does. The statement is about the cutoff M of Eq. (8), not about the coefficients of the Paper 7 condensate; relating the two functionals is a separate open question.*

Conjecture C.1 is promoted below to a conditional theorem (Conditional Theorem C.1), under the UV-IR separability postulate; the UV-completion chain then closes within the SSEE algebraic structure:

$$\varphi \xrightarrow{\alpha=\varphi^4/3} \alpha \xrightarrow{M^4=45\alpha^2\rho_{\rm crit}} M \xrightarrow{s_K^{\rm full}} f_{\rm screen} \xrightarrow{\times H_0^{\rm SH0ES}} H_0^{\rm glob} \;=\; 67.96214\ \mathrm{km/s/Mpc}\,. \tag{28}$$

Conditional on that postulate, the cascade reproduces $3(\varphi+\pi)^2$ to 4.2×10^{-6}. The agreement is a coincidence of central values and not a measurement of the cutoff: the propagated SH0ES uncertainty is ±0.968 km/s/Mpc, which covers M^4 from $0.2\times$ its nominal value upwards. The result is not unconditional.

Diagnosing the Route A discrepancy. Route A matches $K(X)/\mathrm{KAL}_0$ to the Taylor expansion of $K_\alpha(X)/\mathrm{KAL}_0$ and obtains $M_{\rm A}^4 = 6\alpha\,\mathrm{KAL}_0^2 = 2\varphi^4\,\mathrm{KAL}_0^2 \approx 418\,\rho_{\rm crit}$. The discrepancy with the UV value is

$$\frac{M_{\rm A}^4}{M_{\rm UV}^4} \;=\; \frac{2\varphi^4\,\mathrm{KAL}_0^2}{5\varphi^8} \;=\; \frac{2\mathrm{KAL}_0^2}{5\varphi^4} \;=\; \frac{(\varphi+3\pi)^2}{10\varphi^4} \;\approx\; 1.779\,. \tag{29}$$

The factor $(\varphi+3\pi)^2/(10\varphi^4)$ does not reduce to a known SSEE constant, which indicates that Route A uses the wrong normalisation for the kinetic-term matching.

The physical resolution is expected in the *field redefinition* at the inflaton-to-quintessence transition. In quintessential inflation, the canonical inflaton χ is related to ϕ via $\phi = f(\chi)$, where f carries α-attractor factors (e.g., $f(\chi) = \sqrt{6\alpha}\tanh\!\big(\chi/\sqrt{6\alpha}\big)$ for the Starobinsky–Higgs embedding). When expressed in terms of ϕ (the late-time dark energy field), the second-order kinetic coefficient picks up a factor from $|df/d\chi|^2$ evaluated at the transition, which would shift $M_{\rm A}^4$ by $(10\varphi^4)/(\varphi+3\pi)^2 \approx 0.562$ — precisely the inverse of the Route A discrepancy.

UV-IR separability and the conditional derivation. We now show that Conjecture C.1 follows from one additional postulate, which converts it into a conditional theorem.

The α-attractor kinetic Lagrangian is

$$K_\alpha(X) \;=\; -3\alpha\ln\!\left(1-\frac{X}{3\alpha}\right) \;=\; X \;+\; \frac{X^2}{6\alpha} \;+\; \frac{X^3}{27\alpha^2} \;+\; \cdots \tag{30}$$

with $\alpha = \varphi^4/3$. In particular, the second derivative at the origin,

$$\left.\frac{\partial^2 K_\alpha}{\partial X^2}\right|_{X=0} \;=\; \frac{1}{3\alpha}\,, \tag{31}$$

depends *only* on α — and hence only on φ.

The SSEE dark-energy basis is defined by the first-order kinetic coefficient $K_X^{\rm DE}|_0 = 1/\mathrm{KAL}_0$, where $\mathrm{KAL}_0 = (\varphi + 3\pi)/2$ encodes the late-time equation of state. A field-argument rescaling

$$K_{\rm DE}(X) \;=\; K_\alpha\!\left(\frac{X}{\mathrm{KAL}_{\rm eff}}\right) \tag{32}$$

reproduces $K_X^{\rm DE}|_0 = 1/\mathrm{KAL}_{\rm eff}$ and yields a second-order coefficient

$$\left.\frac{\partial^2 K_{\rm DE}}{\partial X^2}\right|_{X=0} \;=\; \frac{1}{\mathrm{KAL}_{\rm eff}^2}\left.\frac{\partial^2 K_\alpha}{\partial X^2}\right|_{X=0} \;=\; \frac{1}{3\alpha\,\mathrm{KAL}_{\rm eff}^2} \;=\; \frac{2}{M^4}\,, \tag{33}$$

so

$$\boxed{M^4 \;=\; 6\alpha\,\mathrm{KAL}_{\rm eff}^2\,.} \tag{34}$$

Route A is the special case $\mathrm{KAL}_{\rm eff} = \mathrm{KAL}_0$, giving $M_{\rm A}^4 = 6\alpha\,\mathrm{KAL}_0^2 \approx 418\,\rho_{\rm crit}$ — too large by the factor $(\varphi + 3\pi)^2/(10\varphi^4) \approx 1.779$ derived in Eq. (29).

No-π separability postulate. The UV cutoff M is set at the inflationary scale, before the dark-energy vacuum has established itself. At that scale, the only algebraic information available is the inflationary α-attractor parameter $\alpha = \varphi^4/3$. The dark-energy constant $\mathrm{KAL}_0 = (\varphi + 3\pi)/2$ — which carries π through the late-time equation of state $w_0 = -\mathrm{AURA}/\Omega$ (the CPL parameterization [29, 30]) — is an *infra-red* quantity absent at the transition.

We therefore postulate:

> *(Separability)* The effective normalisation $\mathrm{KAL}_{\rm eff}$ in Eq. (34) is a polynomial in φ alone; it does not involve π.

Under this postulate, $\mathrm{KAL}_{\rm eff}^2$ must be a polynomial in φ with rational coefficients. The constraint Eq. (34) then reads

$$\mathrm{KAL}_{\rm eff}^2 \;=\; \frac{M^4}{6\alpha} \;=\; \frac{M^4}{2\varphi^4}\,. \tag{35}$$

The unique monomial in φ satisfying both Eq. (35) and the empirically established $M^4 = 5\varphi^8\,\rho_{\rm crit}$ is

$$\mathrm{KAL}_{\rm eff}^2 \;=\; \frac{5\varphi^8}{2\varphi^4} \;=\; \frac{5}{2}\,\varphi^4 \;=\; \frac{15}{2}\,\alpha\,, \qquad \mathrm{KAL}_{\rm eff} \;=\; \varphi^2\sqrt{\tfrac{5}{2}}\,. \tag{36}$$

Note that $\mathrm{KAL}_{\rm eff}^2 = (5/2)\varphi^4\,\rho_{\rm crit}$ is indeed a power of φ times a rational, with no π, confirming the postulate is self-consistent. The factor $\rho_{\rm crit}$ is carried by $\mathrm{KAL}_{\rm eff}^2$ and not by the rational: the dimensionless content of the whole construction is the single number $M^4/(2\varphi^4\rho_{\rm crit}) = 5/2$. Substituting back into Eq. (34):

$$M^4 \;=\; 6\alpha \times \tfrac{15}{2}\,\alpha \;=\; 45\alpha^2\,\rho_{\rm crit} \;=\; 5\varphi^8\,\rho_{\rm crit}\,. \tag{37}$$

This is precisely Conjecture C.1.

The Route A discrepancy is now transparent. Since $M^4 \propto \mathrm{KAL}_{\rm eff}^2$, a ratio r in the normalisation appears as r^2 in the cutoff: $\mathrm{KAL}_0/\mathrm{KAL}_{\rm eff} = 1.333842$ and $(\mathrm{KAL}_0/\mathrm{KAL}_{\rm eff})^2 = 1.779135$, which is exactly the factor by which Route A overshoots ($417.91\,\rho_{\rm crit}$ against $234.89\,\rho_{\rm crit}$). The discrepancy is thus 33% in the normalisation and 78% in the cutoff: the same discrepancy seen at two powers. This ratio is the field-redefinition factor — the "missing" π-dependent normalisation between the inflationary and dark-energy kinematic bases. It is taken in units $\rho_{\rm crit} = 1$; KAL_0 is dimensionless while $\mathrm{KAL}_{\rm eff}$ carries $\rho_{\rm crit}^{1/2}$, so the two are not like-dimensioned objects and 1.333842 must not be read as a comparison between them.

> **Conditional Theorem C.1.** *If the effective kinetic normalisation* $\mathrm{KAL}_{\rm eff}$ *at the inflaton-to-quintessence transition is polynomial in* φ *(the UV-IR separability postulate), then the unique solution to* $M^4 = 6\alpha\,\mathrm{KAL}_{\rm eff}^2$ *consistent with the inflationary* α*-attractor* K_α *is*
>
> $$M^4 \;=\; 45\alpha^2\,\rho_{\rm crit} \;=\; 5\varphi^8\,\rho_{\rm crit}\,. \tag{38}$$
>
> *The UV ladder closes:* $\varphi \to \alpha = \varphi^4/3 \to M^4 = 5\varphi^8\rho_{\rm crit} \to s_K^{\rm full} = 0.41691 \to f_{\rm screen}^{\rm UV} \to H_0^{\rm glob} = 67.96214$ km/s/Mpc *(from the SH0ES datum;* 4.2×10^{-6} *from* $3(\varphi+\pi)^2$*; conditional on Postulate C.1), entirely within the* $\{\varphi, \pi\}$ *algebraic structure of SSEE.*

The unconditional proof requires an explicit derivation of $\mathrm{KAL}_{\rm eff}$ from the $P(X,\phi)$ Lagrangian at the inflationary transition — specifically, showing that the field-space Jacobian $\partial\phi/\partial\chi|_{\rm transition}$ evaluates to $\varphi^2\sqrt{5/2}$ rather than $\mathrm{KAL}_0 = (\varphi + 3\pi)/2$. This is deferred to Paper B of the SSEE series.

7 Additional UV Predictions

Beyond H_0, the UV completion $M^4 = 45\alpha^2\rho_{\rm crit}$ modifies several EFT observables.

Screening rate s_K. The UV-corrected screening rate $s_K^{\rm full} = 0.41691$ is measurable through the equation of state, since $s_K^{\rm IR} = -3w_0(1+w_0)$. Current constraints from DESI DR2 BAO [22] set the background context; forthcoming Euclid and DESI Y5 data will probe w_0, and hence s_K, at the $\sim 1\%$ level, distinguishing the UV-completed theory from the IR limit by $3.4\,\sigma_{\rm syst}$.

Effective equation of state. The UV term X^2/M^4 in $K(X)$ modifies the effective pressure as

$$p_\phi^{\rm UV} = \frac{X}{\mathrm{KAL}_0} + \frac{X^2}{M^4} - V(\phi)\,, \qquad \rho_\phi^{\rm UV} = \frac{X}{\mathrm{KAL}_0} + \frac{3X^2}{M^4} + V(\phi)\,, \tag{39}$$

Under the background matching of Eq. (10) — which holds ρ_ϕ and $\rho_\phi(1+w_0)$ fixed at their IR values — w_0 is unchanged by construction (the field value $X_{\rm bg}$ adjusts instead). A residual shift appears only if one holds the field configuration $(X_{\rm bg}, V)$ fixed when switching on the UV term, in which case the first-order estimate is

$$\Delta w_0^{\rm UV} = w_0^{\rm UV} - w_0^{\rm IR} \approx \frac{(1-3w_0)\,X_{\rm bg}^2/M^4}{\rho_\phi} \approx +2.4\times10^{-3}\,. \tag{40}$$

Either way, $|\Delta w_0| \lesssim 0.002$ lies well below the projected Euclid sensitivity ($\sigma_{w_0} \approx 0.01$): the UV term leaves the background equation of state observationally indistinguishable from the IR limit, and the discriminating observable remains s_K.

Sound speed: one observable, one auxiliary number. The sound speed of the dark-energy field, the one that controls whether dark energy clusters, is that of the Paper 7 condensate, $c_s^2 = (M_v - T_r)/(5M_v + 3T_r) = 0.021284$, fixed by w_0 alone ($c_2X/c_1 = -0.522735$). It is small, so the field barely clusters; the clustering matter of the model is the cold component $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, which is why the growth observables (S_8, $f\sigma_8$) follow the matter density and not the dark-energy field. Paper 5's $c_s^2 \approx 0$ is the Israel–Stewart fluid description of the *same* wave; the field value 0.021284 is the fundamental one (Paper 5; OP-22b). The screening functional of Eq. (8) has its own propagation speed,

$$\frac{K_X}{K_X + 2XK_{XX}} = \frac{1/\mathrm{KAL}_0 + 2X/M^4}{1/\mathrm{KAL}_0 + 6X/M^4} \approx 0.967\,, \tag{41}$$

but Eq. (8) is not the dark-energy action (see the paragraph after Eq. (8)), so this number is a property of the device that transports the cutoff M into s_K, not a prediction for dark-energy perturbations. Whether a single theory furnishes both functionals is the question Postulate C.1 leaves open.

8 Consistency with the SSEE Suite

Table 4 summarises how the UV completion is consistent with the nine preceding papers.

Table 4: Consistency of the UV completion with the SSEE suite.

Paper	Result used	UV impact
Paper 1	$\alpha = \varphi^4/3 \to r = 0.008131$	**Direct input:** $M^4 = 45\alpha^2\rho_{\rm crit}$
Paper 5	IS: $c_s^2 \approx 0$ (fluid description)	Compatible: same wave as Paper 7's field value $c_s^2 = 0.021284$; Eq. (8) is not the DE action
Paper 7	Ghost condensate: w_0, c_s^2 from the Lagrangian	Independent: $s_K^{\rm IR}$ comes from w_0, not from the action
Paper 8	$\mathcal{M} = \mathcal{A}/2$; Vainshtein $r_V \propto 1/M^2$	r_V enlarged to $\sim 10^{44}$ m at UV; double GR protection
Paper 9	$H_0^{\rm glob,IR} = 68.13$ (0.17σ)	UV closes to 67.96214 $(4.3\times10^{-6}\sigma$; conditional[a])
Papers 2–4,6	CMB, MCMC, σ_8, $f\sigma_8$	M only modifies s_K; the algebraic constant $\mathcal{M} = (3\varphi+\pi)/4$ is scale-invariant and unchanged by UV flow

[a] Canonical UV result $H_0^{\rm glob,UV} = 67.96214$ km/s/Mpc, obtained by running the lens on the measured SH0ES rate, conditional on Postulate C.1 (UV-IR separability, §6.3). The propagated uncertainty is ±0.968 km/s/Mpc and dominates the 4.2×10^{-6} residual; the absolute-scale origin (Postulate D) remains open, as in ΛCDM.

The AURA cancellation of Paper 9 ($\mathcal{A}$ drops out of $s_K/(3\mathcal{M})$ in the IR limit) is technically broken in the UV because $s_K^{\rm full}$ acquires an AURA-dependent correction. The canonical UV result

$H_0^{\mathrm{glob,UV}} = 67.96214$ km/s/Mpc, conditional on Postulate C.1, indicates that the UV correction is a consistent completion rather than a destabilisation of the IR identity. As stressed throughout, the absolute dimensional scale (Postulate D) is not an independent prediction — both are self-consistency checks.

9 Summary and Conclusions

We have shown that the SSEE dark-energy Lagrangian $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ admits a unique algebraically consistent UV cutoff

$$M^4 \;=\; 45\,\alpha^2\,\rho_{\mathrm{crit}} \;=\; 5\,\varphi^8\,\rho_{\mathrm{crit}} \;\approx\; 234.9\,\rho_{\mathrm{crit}}\,, \tag{42}$$

where $\alpha = \varphi^4/3$ is the inflationary α-attractor parameter of Paper 1. With this cutoff, the full screening rate is $s_K^{\mathrm{full}} = 0.41691$ and the global rate returned from the SH0ES datum is $H_0^{\mathrm{glob,UV}} = 67.96214$ km/s/Mpc, conditional on Postulate C.1, reproducing $3(\varphi+\pi)^2$ to the seventh digit.

The key conclusions are:

1. The algebraic identity $45\alpha^2 = 5\varphi^8$ (exact, $|\mathrm{diff}| = 0$) connects the inflationary curvature α to the dark-energy UV scale.
2. The UV correction to s_K is perturbative ($\varepsilon \approx 5.7 \times 10^{-4}$) and shifts s_K by $+3.37\%$.
3. The canonical IR Hubble residual of Paper 9 ($H_0^{\mathrm{glob,IR}} = 68.13$, 0.17σ) is reduced to $H_0^{\mathrm{glob,UV}} = 67.96214$ (4.2×10^{-6}) by the UV completion, conditional on Postulate C.1 (§6.3). The cascade does not thereby measure M^4: see §4.
4. The same α predicts $r \approx 0.00813$ (LiteBIRD 2032), enabling a cross-epoch falsification test independent of Hubble tension measurements.
5. A first-principles derivation of $M^4 = 45\alpha^2\rho_{\mathrm{crit}}$ remains under investigation (Section 6), with Route C (full $P(X,\phi)$ Lagrangian) identified as the most promising path.

Acknowledgements. The author thanks Dr. Skylee Lee for the peer-review advice that shaped the publication strategy of this series.

References

[1] M. E. Almeida Vallejo, *SSEE: Algebraic Framework, EFT, and α-Attractor Inflation*, SSEE Paper 1 (2026).

[2] M. E. Almeida Vallejo, *The SSEE Dark-Energy Sector as a Ghost Condensate: Two-Term K-essence, Algebraic Sound Speed, and Bellini-Sawicki Classification*, SSEE Paper 7 (2026).

[3] M. E. Almeida Vallejo, *Strong-Gravity Regime: Disformal Geodesics and $\mathcal{M}$ Lensing*, SSEE Paper 8 (2026).

[4] M. E. Almeida Vallejo, *Hubble Tension via Algebraic Screening Fraction: $f_{\rm screen} = s_K/(3\mathcal{M}) = (\pi-\varphi)/(\varphi+\pi)^2$*, SSEE Paper 9 (2026).

[5] A. G. Riess et al., *A Comprehensive Measurement of the Local Value of the Hubble Constant*, ApJL **934**, L7 (2022) [arXiv:2112.04510].

[6] W. L. Freedman et al., *Status Report on the Chicago-Carnegie Hubble Program (CCHP): Measurement of the Hubble Constant Using the Hubble and James Webb Space Telescopes*, ApJ **985**, 203 (2025) [arXiv:2408.06153v3]. [TRGB, adopted: $H_0 = 70.39 \pm 1.22\,(\text{stat}) \pm 1.33\,(\text{sys}) \pm 0.70\,(\sigma_{\rm SN})$ km/s/Mpc; v1 gave $69.96 \pm 1.05 \pm 1.12$].

[7] E. Bellini & I. Sawicki, *Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity*, JCAP **07** (2014) 050 [arXiv:1404.3713].

[8] LiteBIRD Collaboration, *Probing Cosmic Inflation with the LiteBIRD Cosmic Microwave Background Polarization Survey*, PTEP **2023** (2023) 042F01 [arXiv:2202.02773].

[9] A. A. Starobinsky, *A new type of isotropic cosmological models without singularity*, Phys. Lett. B **91** (1980) 99.

[10] R. Kallosh & A. Linde, *Universality Class in Conformal Inflation*, JCAP **07** (2013) 002 [arXiv:1306.5220].

[11] M. Galante, R. Kallosh, A. Linde & D. Roest, *The Unity of Cosmological Attractors*, Phys. Rev. Lett. **114** (2015) 141302 [arXiv:1412.3797].

[12] A. R. Liddle & S. M. Leach, *How long before the end of inflation were observable perturbations produced?*, Phys. Rev. D **68** (2003) 103503 [arXiv:astro-ph/0305263].

[13] Planck Collaboration (Y. Akrami et al.), *Planck 2018 results. X. Constraints on inflation*, A&A **641**, A10 (2020) [arXiv:1807.06211].

[14] K. N. Abazajian et al. (CMB-S4 Collaboration), *CMB-S4 Science Book, First Edition*, arXiv:1610.02743 (2016).

[15] J. Garriga & V. F. Mukhanov, *Perturbations in k-inflation*, Phys. Lett. B **458** (1999) 219 [arXiv:hep-th/9904176].

[16] G. 't Hooft, *Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking*, in *Recent Developments in Gauge Theories*, NATO ASI Series B **59** (1980) 135.

[17] S. Perlmutter et al., *Measurements of Ω and Λ from 42 high-redshift supernovae*, ApJ **517**, 565 (1999) [arXiv:astro-ph/9812133].

[18] A. G. Riess et al., *Observational evidence from supernovae for an accelerating universe and a cosmological constant*, AJ **116**, 1009 (1998) [arXiv:astro-ph/9805200].

[19] L. Verde, T. Treu & A. G. Riess, *Tensions between the early and the late Universe*, Nat. Astron. **3**, 891 (2019) [arXiv:1907.10625].

[20] M. Kamionkowski & A. G. Riess, *The Hubble tension and early dark energy*, Annu. Rev. Nucl. Part. Sci. **73**, 153 (2023) [arXiv:2211.04492].

[21] Planck Collaboration, *Planck 2018 results VI: Cosmological parameters*, A&A **641**, A6 (2020) [arXiv:1807.06209].

[22] M. Abdul-Karim et al. (DESI Collaboration), *DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints*, Phys. Rev. D **112** (2025) 083515 [arXiv:2503.14738].

[23] E. J. Copeland, M. Sami & S. Tsujikawa, *Dynamics of dark energy*, Int. J. Mod. Phys. D **15**, 1753 (2006) [arXiv:hep-th/0603057].

[24] D. H. Weinberg et al., *Observational probes of cosmic acceleration*, Phys. Rep. **530**, 87 (2013) [arXiv:1201.2434].

[25] G. W. Horndeski, *Second-order scalar-tensor field equations in a four-dimensional space*, Int. J. Theor. Phys. **10**, 363 (1974).

[26] G. Gubitosi, F. Piazza & F. Vernizzi, *The effective field theory of dark energy*, JCAP **2013**, 032 (2013) [arXiv:1210.0201].

[27] C. Armendariz-Picon, V. Mukhanov & P. J. Steinhardt, *Essentials of k-essence*, Phys. Rev. D **63**, 103510 (2001) [arXiv:astro-ph/0006373].

[28] C. Cheung, P. Creminelli, A. L. Fitzpatrick, J. Kaplan & L. Senatore, *The effective field theory of inflation*, JHEP **2008**, 014 (2008) [arXiv:0709.0293].

[29] M. Chevallier & D. Polarski, *Accelerating universes with scaling dark matter*, Int. J. Mod. Phys. D **10**, 213 (2001) [arXiv:gr-qc/0009008].

[30] E. V. Linder, *Exploring the expansion history of the universe*, Phys. Rev. Lett. **90**, 091301 (2003) [arXiv:astro-ph/0208512].

[31] R. R. Caldwell, *A phantom menace? Cosmological consequences of a dark energy component with super-negative equation of state*, Phys. Lett. B **545**, 23 (2002) [arXiv:astro-ph/9908168].

[32] B. Ratra & P. J. E. Peebles, *Cosmological consequences of a rolling homogeneous scalar field*, Phys. Rev. D **37**, 3406 (1988).